

Documentation of New Calculations for Manuscript Revision

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1 Overview

This document records the methodology, parameters, directory organization, and computational rationale for the supplementary calculations performed to address the reviewers' comments for the manuscript "Coordination-Dependent Hydrogen Adsorption on Co-, Fe-, and Ni-Functionalized Monolayer CrCl₃."

The calculations are organized in the `crcl3-newcals` directory. Three complementary suites are established:

1. **DFT-D3 (van der Waals) Dispersion Corrections** (`DFT_D3/`)
2. **DFT+U Hubbard Benchmarks** (`DFT_U/`)
3. **Vibrational Frequency Analysis** (`Freq/`)

2 Methodology and Computational Details

2.1 1. Extraction of Adsorption Energy (ΔE_{ads}) and Clean Substrates

To rigorously extract the hydrogen adsorption energy:

$$\Delta E_{\text{ads}} = E(\text{doped-CrCl}_3 + \text{H}) - E(\text{doped-CrCl}_3) - \frac{1}{2}E(\text{H}_2) \quad (1)$$

both the hydrogenated slab ($E(\text{doped-CrCl}_3 + \text{H})$) and the corresponding clean substrate ($E(\text{doped-CrCl}_3)$) must be evaluated at the exact same level of theory. Therefore, for both **DFT-D3** and **DFT+U**, separate calculations are staged for:

- The 6 hydrogenated systems: `adsorbed/{Co, Fe, Ni}` and `embedded/{Co, Fe, Ni}`.
- The 6 clean substrates without H: `clean/adsorbed/{Co, Fe, Ni}` and `clean/embedded/{Co, Fe, Ni}`.

This guarantees consistency, preventing spurious energetic offsets that would arise from mixing plain PBE and corrected total energies.

2.2 2. DFT-D3 (van der Waals) Setup

Goal: Address Reviewer 4 (Comment 2) by confirming that long-range dispersion does not alter the relative adsorption trends ($\text{Co} < \text{Fe} < \text{Ni}$) or structural coordination stability.

- **Method:** Grimme D3 zero-damping scheme (`IVDW = 11`).
- **Ionic Relaxation:** Full relaxation with `IBRION = 2`, `NSW = 500`, `ISIF = 2`, `EDIFFG = -0.025 eV/Å`.
- **Spin Polarization:** `ISPIN = 2` with dopant-specific initial magnetic moments ($3.0 \mu_{\text{B}}$ for Co, $4.0 \mu_{\text{B}}$ for Fe, $2.0 \mu_{\text{B}}$ for Ni, $3.0 \mu_{\text{B}}$ for Cr).

2.3 3. DFT+U Hubbard Benchmarks

Goal: Address Reviewer 4 (Comments 1 & 16) by verifying that the 3d electron correlation on Cr does not affect the catalytic ordering governed by TM orbital hybridization.

- **Method:** Rotationally invariant formulation by Dudarev et al. (LDAUTYPE = 2) with $U_{\text{eff}} = 4.0$ eV applied exclusively to Cr 3d states.

- **INCAR Specifications:**

```
LDAU      = .TRUE.
LDAUTYPE  = 2
LDAUL     = 2 -1 -1 -1    # (2 -1 -1 for clean substrates)
LDAUU     = 4.0 0 0 0    # (4.0 0 0 for clean substrates)
LDAUJ     = 0 0 0 0      # (0 0 0 for clean substrates)
LMAXMIX   = 4            # Mandatory for d-state PAW charge density mixing
```

- **Critical Verification:** LMAXMIX = 4 is explicitly included to avoid charge mixing truncation at $l = 2$.

2.4 4. Vibrational Frequency Calculations

Goal: Address Reviewer 4 (Comment 4) by obtaining explicit zero-point energy (ΔE_{ZPE}) and entropic ($-T\Delta S$) corrections for the adsorbed hydrogen.

- **Method:** Finite difference of forces (IBRION = 5, NFREE = 2, displacement POTIM = 0.015 Å).
- **Precision:** Tight electronic convergence EDIFF = 1.0e-07 to eliminate force noise.
- **Constraints:** Selective dynamics in POSCAR. Substrate atoms 1–33 (Cr, Cl, TM) are fixed (F F F), while the adsorbed H atom (atom 34) is allowed to vibrate (T T T). Starting coordinates are initialized directly from fully converged CONTCAR geometries.

3 Summary of Staged Calculations

A total of **30 distinct calculations** are staged and ready for cluster execution:

Suite	Configuration	Systems	Total Jobs
DFT_D3	Hydrogenated (adsorbed + embedded)	Co, Fe, Ni	6
DFT_D3	Clean Substrates (adsorbed + embedded)	Co, Fe, Ni	6
DFT_U	Hydrogenated (adsorbed + embedded)	Co, Fe, Ni	6
DFT_U	Clean Substrates (adsorbed + embedded)	Co, Fe, Ni	6
Freq	Hydrogenated Selective Dynamics	Co, Fe, Ni	6
Total			30

4 Cluster Deployment and Automation

Execution on the HPC cluster is automated through:

- DFT_D3/run_monitor_vasp.sh

- `DFT_U/run_monitor_vasp.sh`
- `Freq/run_monitor_vasp.sh`
- `run_all_new_calculations.sh` (root master orchestrator)

These scripts check file completeness, track active SLURM jobs to avoid duplication, and assign unique job identifiers (e.g., `D3_ads_Co`, `U_cln_emb_Fe`, `Freq_ads_Ni`).